In [1]:
```
# importing major libraries

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:
```
import warnings
warnings.filterwarnings('ignore')
```

In [3]:
```
df=pd.read_csv('loan_approval_dataset.csv').drop(columns='loan_id')
```

In [4]:
```
df.head()
```

Out[4]:

	no_of_dependents	education	self_employed	income_annum	loan_amount	l
0	2	Graduate	No	9600000	29900000	
1	0	Not Graduate	Yes	4100000	12200000	
2	3	Graduate	No	9100000	29700000	
3	3	Graduate	No	8200000	30700000	
4	5	Not Graduate	Yes	9800000	24200000	

In [5]:
```
df.shape
```

Out[5]:
```
(4269, 12)
```

In [6]:
```
#df.loan_status
df.columns.str.strip()
```

Out[6]:
```
Index(['no_of_dependents', 'education', 'self_employed', 'income_annum',
       'loan_amount', 'loan_term', 'cibil_score', 'residential_assets_value',
       'commercial_assets_value', 'luxury_assets_value', 'bank_asset_value',
       'loan_status'],
      dtype='object')
```

In [7]:
```
df.columns=df.columns.str.strip()
```

In [8]:
```
df.columns
```

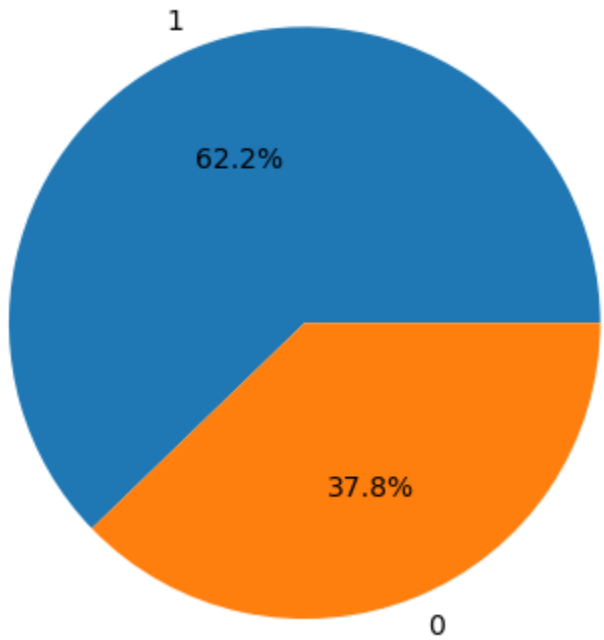
Out[8]:
```
Index(['no_of_dependents', 'education', 'self_employed', 'income_annum',
       'loan_amount', 'loan_term', 'cibil_score', 'residential_assets_value',
       'commercial_assets_value', 'luxury_assets_value', 'bank_asset_value',
       'loan_status'],
      dtype='object')
```

In [9]:
```
df.loan_status.unique()
```

Out[9]: array([' Approved', ' Rejected'], dtype=object)

In [10]:
```
df.loan_status=df.loan_status.map({' Approved':1, ' Rejected':0})
```

In [11]:
```
plt.pie(df.loan_status.value_counts(),labels=df.loan_status.value_counts().inc
plt.show()
df.loan_status.value_counts()
```



Out[11]:
```
loan_status
1    2656
0    1613
Name: count, dtype: int64
```

In [12]:
```
df.education.unique()

df.education=df.education.str.strip()
df.self_employed=df.self_employed.str.strip()
```

In [13]:
```
# column transformer
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder
l1 = ['Graduate','Not Graduate']
l2  =['No', 'Yes']
trans1 = ColumnTransformer([
    ('tr1',OneHotEncoder(categories=[l1],drop='first',dtype='int'),['education
    ('tr2',OneHotEncoder(categories=[l2],drop='first',dtype='int'),['self_empl
],remainder='passthrough')
trans1
```

Out[13]:

```
▸ ColumnTransformer
▸ tr1          ▸ tr2          ▸ remainder
  ▸ OneHotEncoder   ▸ OneHotEncoder   ▸ passthrough
```

In [14]:

```
#df.education.unique()
#df.self_employed.unique()
```

In [15]:

```
x=df.iloc[:,0:-1]
y=df.loan_status


#train test split
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test = train_test_split(x,y,test_size=0.2,random_stat
```

In [16]:

```
from sklearn.ensemble import RandomForestClassifier
rdc = RandomForestClassifier(max_depth=3,n_jobs=-1,n_estimators=101)
rdc
```

Out[16]:

```
▾ RandomForestClassifier
RandomForestClassifier(max_depth=3, n_estimators=101, n_jobs=-1)
```

In [17]:

```
# fit
x_train = pd.DataFrame(trans1.fit_transform(x_train),columns=trans1.get_featur
x_test = pd.DataFrame(trans1.fit_transform(x_test),columns=trans1.get_feature_
```

In [18]:

```
rdc.fit(x_train,y_train)
```

Out[18]:

```
▾ RandomForestClassifier
RandomForestClassifier(max_depth=3, n_estimators=101, n_jobs=-1)
```

In [19]:

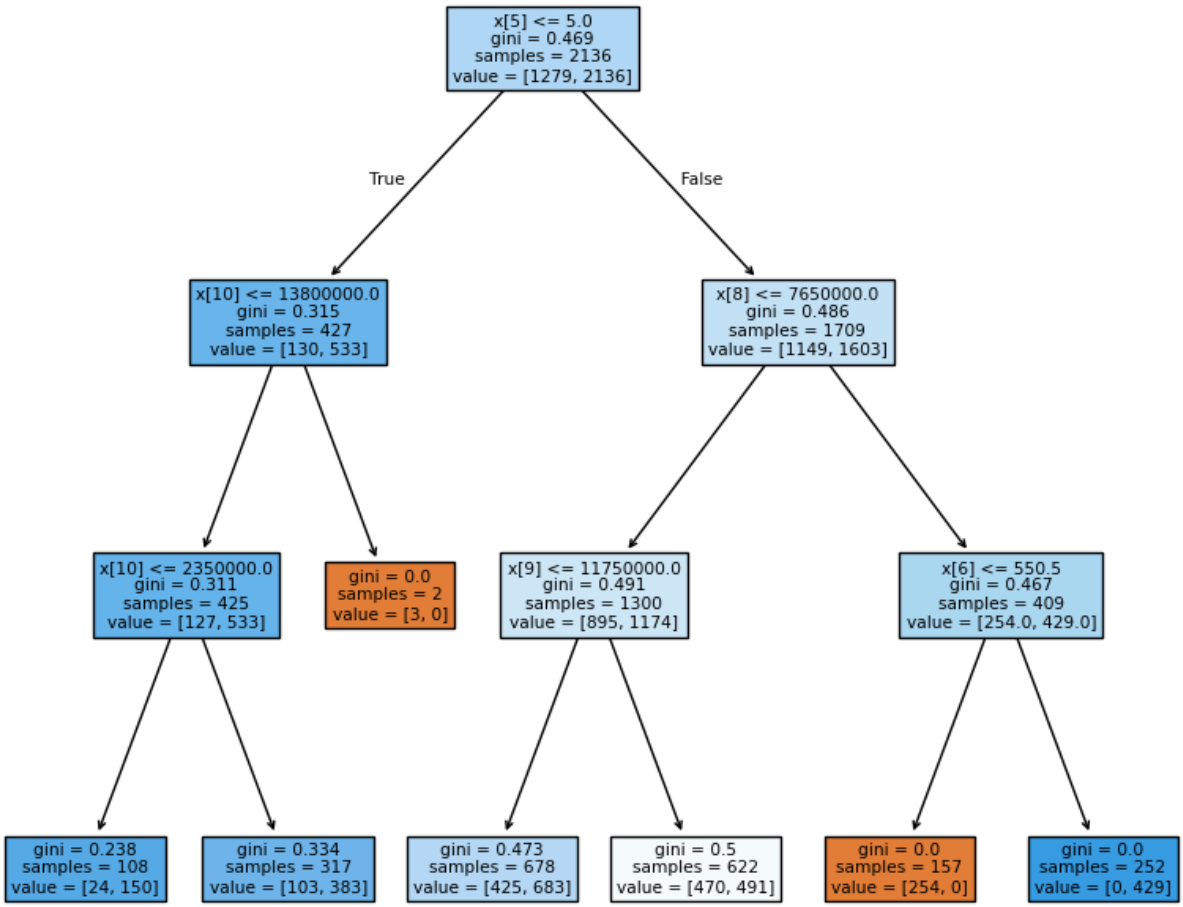
```
y_pred = rdc.predict(x_test)

from  sklearn.metrics import accuracy_score,precision_score,f1_score

print('accuracy',accuracy_score(y_test,y_pred))
print('precision',precision_score(y_test,y_pred))
print('f1_score',f1_score(y_test,y_pred))
```

```
accuracy 0.9590163934426229
precision 0.9980119284294234
f1_score 0.9663137632338787
```

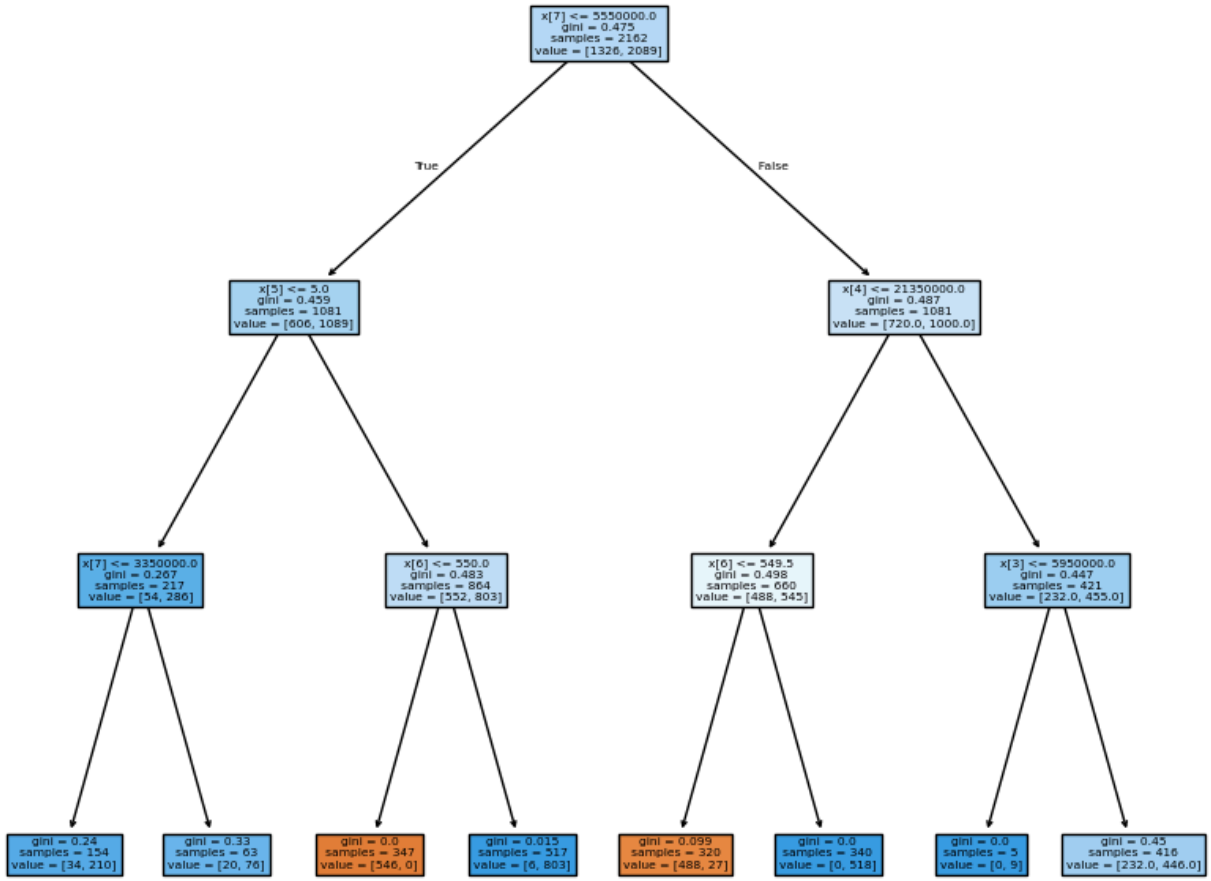
In [20]:
```
#fitting
rdc.score(x_train,y_train)
```

Out[20]: 0.9592972181551976

In [21]:
```
from sklearn import tree
for i in range(2):
    plt.figure(figsize=(10,9))
    tree.plot_tree(rdc[i],filled=True)
    plt.show()
    print('-'*120)
```




```
------------------------------------------------------------------------------------------------------------------------
```



In [22]:
```
rdc.feature_importances_
```

Out[22]:
```
array([9.79050900e-05, 1.19428269e-03, 2.35442730e-03, 1.15087874e-02,
       1.79283664e-02, 1.32685216e-01, 7.94948119e-01, 1.08913036e-02,
       7.12170191e-03, 9.59546905e-03, 1.16744212e-02])
```

In [23]:
```
temp = pd.Series(rdc.feature_importances_,index=x_train.columns).reset_index()
```

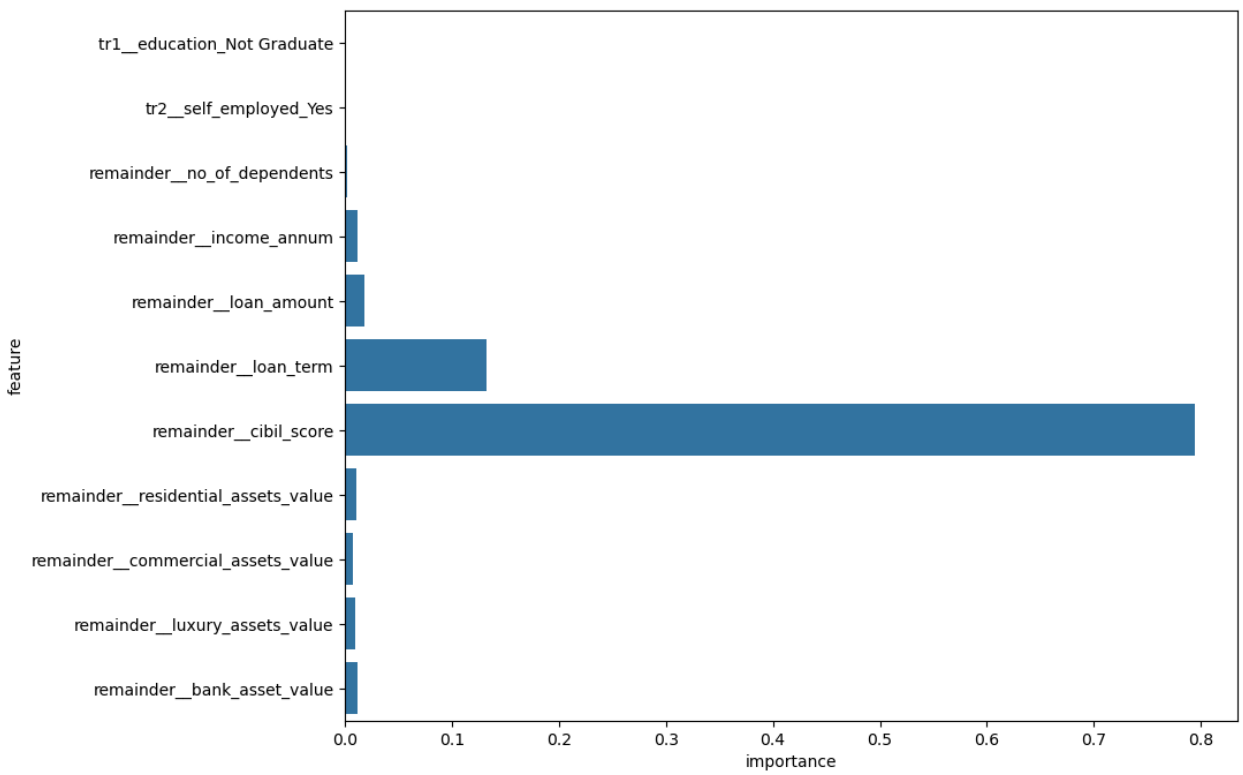
In [24]:
```
temp.columns = ['feature','importance']
temp
```

Out[24]:

	feature	importance
0	tr1__education_Not Graduate	0.000098
1	tr2__self_employed_Yes	0.001194
2	remainder__no_of_dependents	0.002354
3	remainder__income_annum	0.011509
4	remainder__loan_amount	0.017928
5	remainder__loan_term	0.132685
6	remainder__cibil_score	0.794948
7	remainder__residential_assets_value	0.010891
8	remainder__commercial_assets_value	0.007122
9	remainder__luxury_assets_value	0.009595
10	remainder__bank_asset_value	0.011674

In [25]:

```
plt.figure(figsize=(10,8))
sns.barplot(data=temp,x='importance',y='feature')
plt.show()
```



In [ ]: